LESSON 7.4

GRAPHING LINES FROM SLOPE-INTERCEPT FORM

LESSON INTRODUCTION

MA.8.AR.3.4 Given a mathematical or real-world context, graph a two-variable linear equation from a written description, a table, or an equation in slope-intercept form.

LEARNING TARGETS

- I can graph a two-variable linear equation from an equation in slope-intercept form.
- I can graph a two-variable linear equation from a table of values.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.4.3 Given a mathematical or real-world context, graph proportional relationships from a table, equation, or a written description.--

WORKING TOWARD

MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function and determine and interpret its key features.

ESSENTIAL QUESTIONS

- How can you use the slope and y-intercept of a line to graph a linear relationship?
- How many points do you need to graph a line? Does it matter which two points you use?
- How can you use the given equation to find two points on the line?

LESSON NARRATIVE

In this lesson, students apply what they learned in the first three lessons about representations of linear relationships to graph the relationships on a coordinate plane from given equations. Students will explore how linear relationships can be graphed using either the slope and y-intercept or two points on the line. Students will interact with linear equations in both slope-intercept form and standard form. When given an equation in standard form, they will use it to find two points on the line to create the graph.

COMPONENT SUMMARY

7.4.1 WARM-UP *(~5 MIN.)*

Given a point, determine a point that will create the steepest possible line

7.4.2 GUIDED INSTRUCTION *(10-15 MIN.)*

Learn how to graph two-variable linear equations from equations and tables of values

7.4.3 TRY IT *(10 MIN.)*

Practice graphing two-variable linear equations

7.4.4 COLLABORATION *(5-10 MIN.)*

Analyze an error a student made graphing a two-variable linear equation

7.4.5 YOUR TURN! *(10 MIN.)*

Practice graphing two-variable linear equations

7.4.6 WRAP-UP *(~5 MIN.)*

Demonstrate understanding of graphing two-variable linear equations

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Slope-intercept form
- Linear equation
- Two-variable linear equation
- Slope
- y-intercept

MATERIALS

- Graphing calculators or Desmos
- Straightedge

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle determining what information to use from the equation or table to graph the linear relationship.
- Students may find and plot multiple points when only two are needed.
- Students may try to sketch the graph instead of using points and a straightedge.
- Students may struggle finding points using the equation.

THINGS TO CONSIDER

- Create an anchor chart with instructions and examples of how to graph a linear relationship from a table and equation.
- Use the word “substitution” rather than “plugging in” when students are using the equation to find points on the graph.
- If time is an issue, consider assigning 7.4.5 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

Consider using the Think-Pair-Share routine in the 7.4.4 Collaboration. Giving students time to work independently prior to working with their partner will lead to richer discussions. The discussions will build student confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In 7.4.4 Collaboration, students analyze and critique a sample of student work. Students are less likely to make the mistake they identify themselves in the future. Provide students with sentence starters if they struggle to explain their thinking.

DIFFERENTIATE INSTRUCTION

Consider allowing students to create reference cards in 7.4.2 Guided Instruction and use them in 7.4.3 Try It and 7.4.5 Your Turn!. Have students include instructions and examples on their cards. Consider allowing students to use Desmos or graphing calculators to check their answers. The graphs throughout the lesson provide visual entry points. The discussions in 7.4.4 Collaboration provide auditory entry points. In the 7.4.6 Wrap-Up, students are asked to provide written explanations. If students struggle with writing an explanation, consider allowing them to speak their responses into an audio recorder.

7.4.1 WARM-UP: MAXIMIZING SLOPE

TEACHER MOVES

If time allows, have students discuss their strategies with a partner. Ideally, the discussion leads to a developing understanding of the ratio between each set of numbers being important. Consider having students use Desmos or another graphing utility to experiment with the ordered pairs.

QUESTIONING

To add complexity, consider challenging the students by expanding the range of numbers they can choose from to include negatives (for example, a range from -9 to 9).

7.4.1 Warm-Up: Maximizing Slope

1. Place a digit in each blue box to find a point that creates the steepest slope for a line that passes through the point $(3, 5)$. Use only digits 1 to 9, at most one time each.

$(\square, \square)$

Answers will vary. Both $(4,9)$ and $(2,1)$ would produce the steepest slope.

7.4.2 Guided Instruction: Graphing Two-Variable Linear Equations

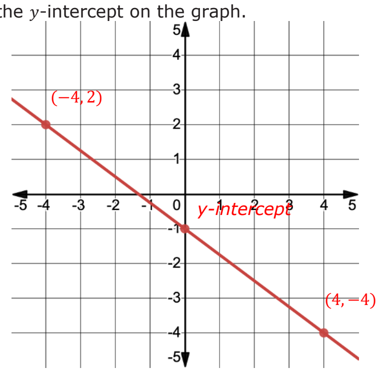
The graph of a linear relationship can be created from a two-variable linear equation.
When given the equation in slope-intercept form, start by determining the slope and the y-intercept of the line.

1. Complete the table.

Equation	Slope	y-intercept
$y = -\frac{3}{4}x - 1$	$-\frac{3}{4}$	-1

Recall that the y-intercept is the value of y when $x = 0$, at the point where the line intersects the y-axis.

2. Write the y-intercept as a coordinate pair. $(\ 0\ ,\ -1\ )$
3. Plot the y-intercept on the graph.



4. Use the slope to find two other points on the line.
   $(-4,2)$ and $(4,-4)$ plotted on graph.
5. Use a straightedge to draw the line that represents the linear relationship.
   Shown on graph.

Two-variable linear relationships can also be written in other forms. When given the equation in another form, start graphing by determining points on the line using the equation.

7.4.2 GUIDED INSTRUCTION: GRAPHING TWO-VARIABLE LINEAR EQUATIONS

ENRICHMENT

- How can you determine the slope from the equation in slope-intercept form?
- How can you determine the y-intercept from the equation in slope-intercept form?
- Which value represents a specific point on the graph of the line?

7.4.2 GUIDED INSTRUCTION: GRAPHING TWO-VARIABLE LINEAR EQUATIONS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Students are not expected to be able to recognize forms of lines other than slope-intercept form. Model for students how to create a table of values when given equations in other forms. Discuss why it is helpful to use the intercepts in this case to determine points to easily identify on the line.

TEACHER MOVES

Teachers should listen to student discussion on question 8 to observe whether students understand the slope of a line between any two points is always the same. If student discussions indicate a misunderstanding of this concept, consider facilitating a whole-group discussion with examples using the line shown.

ENRICHMENT

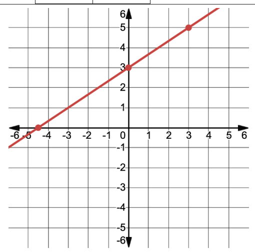
Consider asking students to paraphrase this paragraph in the student workbook using the following prompt:

"Read the following paragraph (in the workbook, seen here) and restate in your own words. Provide an example."

6. Complete the table of values using the equation.

Equation	Table of Values x	Table of Values y
$-2x + 3y = 9$	-4.5	0
	0	3
	3	5

7. Plot the ordered pairs from the table of values on the graph. Then, use a straight-edge to draw the line that represents the linear relationship.
Shown on graph.



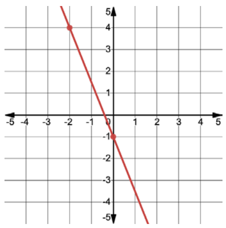
8. Ana says that to graph a linear relationship, only two points on the line are needed. Discuss with your partner whether you agree or disagree with Ana and explain why.

9. Complete the statements.
The slope of the line between any two points in a linear relationship ○ varies ○ is the same because the rate of change, $\frac{\text{change in } y}{\text{change in } x}$, is ○ constant. ○ always changing.
Therefore, ○ only two points ○ more than two points are needed to graph the line using a straightedge.

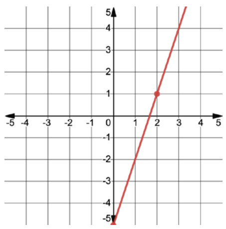
When given the equation of a line in slope-intercept form, two points on the line can be determined using the y-intercept and slope. Start by graphing the y-intercept. Then, use the slope to determine a second point to graph from there. When given the equation of a line in a form other than slope-intercept, two points on the line can be determined by using substitution to create a table of values.

7.4.3 Try It: Graphing Two-Variable Linear Equations

1. Graph the equation $y = -\frac{5}{2}x - 1$.



2. Graph the equation $y = 3x - 5$.

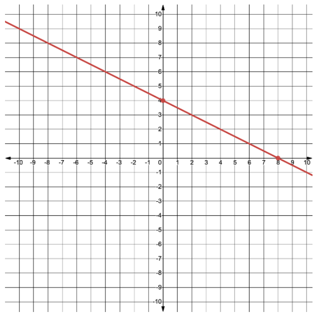


3. Consider the equation $x + 2y = 8$.

   a. Use the equation to complete the table of values.
   Answers will vary.

x	y
0	4
8	0

   b. Use the table of values to graph the equation.



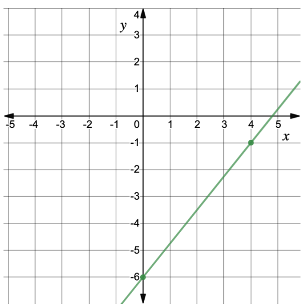
7.4.3 TRY IT: GRAPHING TWO-VARIABLE LINEAR EQUATIONS

TEACHER MOVES

Consider completing question 1 as a whole group to solidify understanding prior to giving the students the opportunity to complete the remainder of the questions either independently or with a partner.

7.4.4 Collaboration: Error Analysis

Nico graphed the equation $y = \frac{4}{5}x - 6$. His graph is shown.



1. Nico made an error when he graphed the line $y = \frac{4}{5}x - 6$. Work with your partner to find the error.

2. Help Nico understand how he can learn from his mistake. Write a brief note to Nico explaining what he did correctly, what his mistake was, and how he can learn from it.

   Answers will vary. Example: Nico, it looks like you correctly identified and plotted the y-intercept at $(0, -6)$. However, you did not find a second point on the line correctly. The slope of the line is $\frac{4}{5}$. It looks like you plotted a point that is 5 units up and 4 units right of the y-intercept when you should have gone 4 units up and 5 units to the right for the second point. Remember, slope is the change in y (vertical distance) over the change in x (horizontal distance), not the other way around.

7.4.4 COLLABORATION: ERROR ANALYSIS

TEACHER MOVES

Listen to student discussions of the error Nico made to determine if they are recognizing that he correctly plotted the y-intercept but did not correctly find a second point. If students are not seeing the error, consider asking the following questions to probe thinking:

- Did Nico plot the y-intercept correctly?
- What other point did Nico use to create the line, and how did he determine its location?

QUESTIONING

Have students discuss other possible errors that could be made when graphing.

7.4.5 YOUR TURN!

ENRICHMENT

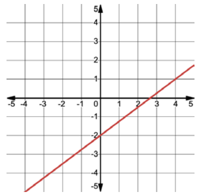
Based on the graph provided, what are some x-values that could be used to graph 5x - 2y = 6? Name at least two different ways to determine the slope of a line.

TEACHER MOVES

Struggling students may benefit from hearing from others regarding why they chose specific values for their tables. Ask students to share their logic and reasoning.

7.4.5 Your Turn!

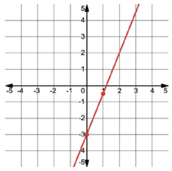
1. Graph the equation $y = \frac{3}{4}x - 2$.



2. Consider the equation $5x - 2y = 6$.
   a. Use the equation to complete the table of values.
   Answers will vary.

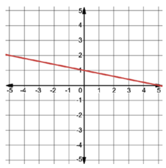
x	y
0	-3
1	$-\frac{1}{2}$

   b. Use the table of values to graph the equation $5x - 2y = 6$.



   c. What is the slope of the line?
   The slope is $\frac{5}{2}$.

3. Graph the equation $y = -\frac{1}{5}x + 1$.



7.4.6 Wrap-Up: Reflection

1. Complete the learning pyramid based on today's lesson.
   Answers will vary.

One question I would like to have answered …	
One mistake I learned from today …	One thing I am not sure about … YET!
One thing I am confident I know is …	

7.4.6 WRAP-UP: REFLECTION

TEACHER MOVES

Discuss any questions the students still have regarding the content. Consider a "parking lot" approach, with a list of the questions on the board or poster in which students can either answer each other's questions (perhaps for bonus points) or you can revisit as a class in later lessons.

